

Baseline Analysis Brief

GMMNet — *Synthesis of Electric Vehicle Charging Data: A Real-World Data-Driven Approach*

Z. Li, Z. Bian, Z. Chen, K. Ozbay, M. Zhong — *Communications in Transportation Research* 4(3):100128, 2024 (open access, CC BY; Tsinghua University Press). NYU Shanghai / NYU C2SMARTER / Shanghai EV Public Data Center. DOI 10.1016/j.commtr.2024.100128.

1. What the paper does

The paper introduces a data-driven generator that learns the joint distribution of EV charging events and synthesises new, realistic events — motivated by the scarcity, privacy restrictions and cost of real large-scale charging data. It is trained and validated on a real-world dataset of approximately **1.65 million charging events from 3,777 battery EVs (BEVs) in Shanghai** over one year. The learned model reproduces the underlying distribution of the original data and supports **data imputation, augmentation, and conditional generation** of future charging distributions under anticipated development premises. It is referred to as “GMMNet” in subsequent literature (e.g. the 2026 copula-vs-neural study), where it serves as the state-of-the-art neural benchmark.

2. Method

- **Conditional density networks.** A set of mixture-density neural networks, one per variable, that output the parameters of a Gaussian Mixture Model for each charging-event feature *conditioned on the other features* (rather than a single unconditional joint density). As reimplemented in the benchmark literature: two 32-unit hidden layers with batch normalisation and ReLU, an output layer predicting a 5-component GMM, trained by minimising the negative log-likelihood (Adam).
- **Gibbs sampling for generation.** Because the model provides full conditionals, new joint samples are drawn by Gibbs sampling — iteratively resampling each variable from its conditional density given the current values of the others until the chain mixes. This yields synthetic charging events that preserve inter-feature dependencies.
- **Conditional / scenario generation.** By conditioning the networks on chosen premises (e.g. anticipated fleet or infrastructure development), the generator can produce forward-looking charging distributions, not just replicate the training distribution.

3. Data & evaluation

Trained on ~1.65M events / 3,777 BEVs (Shanghai, one year) — a large fleet-telemetry regime. The evaluation shows the generated samples closely resemble real charging events and that the model captures the underlying joint distribution; the authors demonstrate downstream utility for imputation and augmentation. In the independent 2026 copula-vs-neural comparison, GMMNet is a strong baseline (notably competitive on the Trondheim / Norway data), but joint-dependency models (Vine copula and the CODINE neural copula) matched or outperformed it on preserving *tail dependence* and *rank-correlation* structure.

4. Relevance and positioning in our study

Aspect	GMMNet (Li et al. 2024)
Class	Neural, conditional mixture-density + Gibbs sampling
Output	Session/event-level joint (data-fitted) — compatible with our (arrival, duration, energy)
Comparable?	Yes — data-fitted; fits our “fit → generate → compare marginals” protocol
Baseline role	Candidate for the neural slot ; also the reference neural benchmark in recent EV copula work
Availability	No official public repo located; implementable in PyTorch per the architecture above

How it contrasts with our scheme. GMMNet models the joint via *conditional* mixture-density networks over pooled fleet data and samples with Gibbs; it does not model *per-user* heterogeneity or the sequential idle-gap / sojourn structure. Our per-user semi-Markov chain fits each user’s transition matrix and sojourn distributions explicitly. GMMNet is therefore an ideal *neural* baseline: it is domain-specific, dependency-aware, and recent — and beating it (together

with a copula baseline and EV-SDG) directly substantiates the value of per-user modelling. A quantitative head-to-head on our datasets is pending implementation.